

## 4.12.13

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# Question

If the equation of the base of an equilateral triangle is

$$x + y = 2, \quad \text{and the vertex is } (2, -1),$$

find the length of the side of the triangle and the coordinates of the other two vertices.

# Step 1: Vector Form

Equation of base line:

$$\vec{n}^T \vec{x} = 2 \quad (1)$$

where  $\vec{n} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ .

Vertex:

$$\vec{A} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}.$$

## Step 2: Foot of Perpendicular (Matrix Form)

The foot  $\vec{D}$  is obtained by:

$$\vec{D} = \vec{A} - \frac{\vec{n}^T \vec{A} - 2}{\vec{n}^T \vec{n}} \vec{n}.$$

Substitute values:

$$\vec{D} = \begin{pmatrix} 2 \\ -1 \end{pmatrix} - \frac{2 - 1 - 2}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{5}{2} \\ -\frac{1}{2} \end{pmatrix}.$$

## Step 3: Altitude Relation

Altitude length:

$$AD = \|\vec{A} - \vec{D}\| = \frac{1}{\sqrt{2}}.$$

Equilateral triangle relation:

$$AD = \frac{\sqrt{3}}{2}a \quad \implies \quad a = \sqrt{\frac{2}{3}}.$$

## Step 4: Other Vertices

Direction vector along base:

$$\vec{m} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad \|\vec{m}\| = \sqrt{2}.$$

Vertices of base:

$$\vec{B} = \vec{D} + \frac{a}{2} \frac{\vec{m}}{\|\vec{m}\|}, \quad \vec{C} = \vec{D} - \frac{a}{2} \frac{\vec{m}}{\|\vec{m}\|}.$$

Final:

$$B = \left( \frac{5}{2} + \frac{1}{2\sqrt{3}}, -\frac{1}{2} - \frac{1}{2\sqrt{3}} \right), \quad C = \left( \frac{5}{2} - \frac{1}{2\sqrt{3}}, -\frac{1}{2} + \frac{1}{2\sqrt{3}} \right).$$

# Python Code

```
import numpy as np
import matplotlib.pyplot as plt

A = np.array([2, -1])      # Vertex
n = np.array([1, 1])      # Normal to base
# Foot of perpendicular
D = A - ((n @ A - 2) / (n @ n)) * n

# Side length
a = np.sqrt(2/3)

# Base direction vector
m = np.array([1, -1]) / np.sqrt(2)

# Other vertices
B = D + (a/2)*m
C = D - (a/2)*m
```

# Python Plot Code

```
x = np.linspace(-2, 4, 400)
y = 2 - x

plt.plot(x, y, 'k-', label=r"$x+y=2$")

# Triangle
plt.plot([A[0], B[0], C[0], A[0]],
         [A[1], B[1], C[1], A[1]], 'b-')

# Altitude
plt.plot([A[0], D[0]], [A[1], D[1]], 'r--')

# Points
for P, name in [(A, "A"), (B, "B"), (C, "C"), (D, "D")]:
    plt.scatter(*P)
    plt.text(P[0]+0.1, P[1], name, fontsize=10)

plt.gca().set_aspect('equal')
plt.grid(True, linestyle='--', alpha=0.6)
```



# C Code

```
#include <stdio.h>
#include <math.h>

int main() {
    double Ax=2, Ay=-1;
    double Dx=2.5, Dy=-0.5;

    double AD = sqrt((Ax-Dx)*(Ax-Dx) + (Ay-Dy)*(Ay-Dy));
    double a = (2 * AD) / sqrt(3);

    printf("Side length = %lf\n", a);

    // Base direction vector normalized
    double mx=1/sqrt(2), my=-1/sqrt(2);

    double Bx = Dx + (a/2)*mx;
    double By = Dy + (a/2)*my;
    double Cx = Dx - (a/2)*mx;
```

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